FINDING THE RANGE

Which method for the range?

How the function is presented, and what is already known about it, picks the method.

1 Is the graph given? — YES → **Read it from the graph**
trace the vertical extent; check solid and open dots, and asymptotes → Range from a graph

NO

2 An applied or physical problem? — YES → **Constrain by meaning**
height ≥ 0, probability in [0, 1], and so on → Range from context
a cost with a 500-dollar start-up: the range begins at 500

NO

3 Is the inverse known or easy to find? — YES → **Swap with the inverse**
the range of f is the domain of f^{-1} → Range and invertibility
e^x: range = domain of ln = (0, ∞)

NO

4 Does it match a common parent?
possibly shifted or stretched — YES → **Look up the family**
take the parent's range, then adjust for the transformations → Common function types
a^x has range (0, ∞); shifting up by k gives (k, ∞)

NO

5 Can y = f(x) be solved for x?
rational, quadratic, simple radical forms

YES → **Solve for x**
the y-values that give a real x form the range → Finding range algebraically
$y = x^2 \to x = \pm\sqrt{y}$, so $y \geq 0$

NO → **Analyze the behavior**
find the minimum and maximum; check end limits and asymptotes → Analyzing behavior